Clustering With K Means

```
from sklearn.cluster import KMeans
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from matplotlib import pyplot as plt
%matplotlib inline
```

```
df = pd.read_csv("income.csv")
df.head()
```

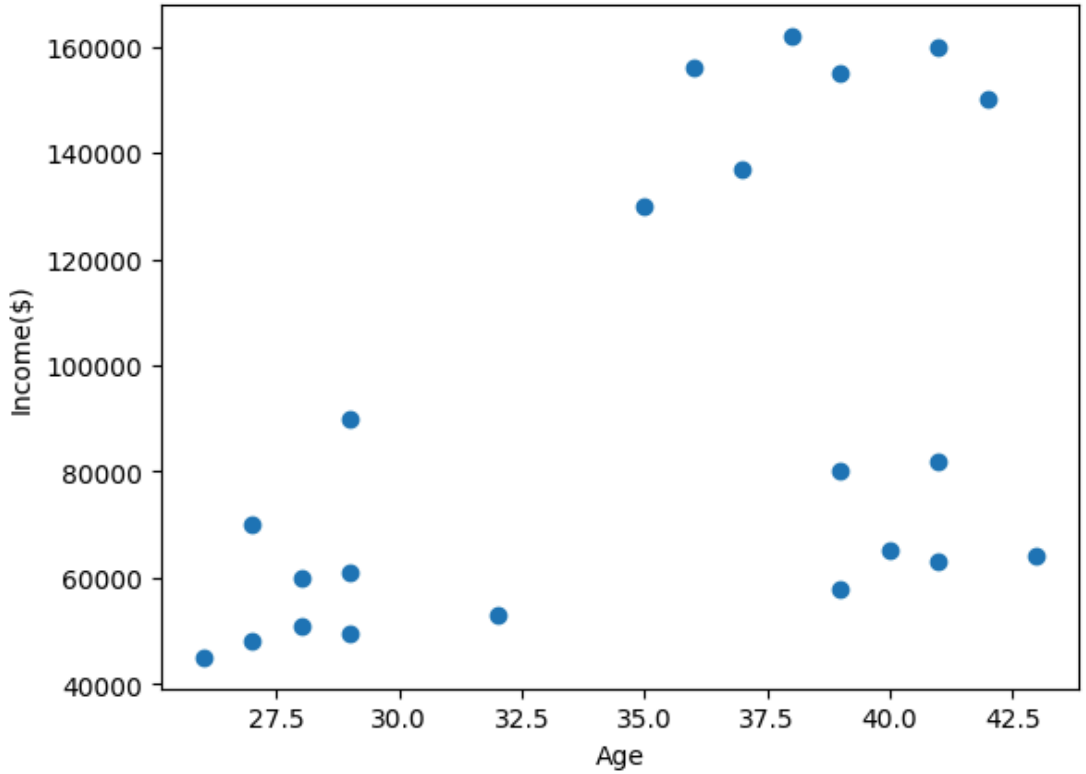
	Name	Age	Income($)
0	Rob	27	70000
1	Michael	29	90000
2	Mohan	29	61000
3	Ismail	28	60000

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EDA

```
plt.scatter(df.Age,df['Income($)'])
plt.xlabel('Age')
plt.ylabel('Income($)')
```

```
Text(0, 0.5, 'Income($)')
```



KMeans

```
km = KMeans(n_clusters=3,init = 'k-means++')
y_predicted = km.fit_predict(df[['Age','Income($)']])
y_predicted
```

```
array([2, 0, 2, 2, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 0, 0, 2],
      dtype=int32)
```

Clustering Output Visualization

```
df['cluster']=y_predicted
df.head()
```

	Name	Age	Income($)	cluster
0	Rob	27	70000	2
1	Michael	29	90000	0
2	Mohan	29	61000	2
3	Ismail	28	60000	2
4	Kory	42	150000	1

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```
df['cluster']= km.labels_
df.head()
```

	Name	Age	Income($)	cluster
0	Rob	27	70000	2
1	Michael	29	90000	0
2	Mohan	29	61000	2
3	Ismail	28	60000	2
4	Kory	42	150000	1

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```
df.groupby('cluster').describe().T
```

	cluster	0	1	2
Age	**count**	3.000000	7.000000	12.000000
	mean	36.333333	38.285714	32.416667
	std	6.429101	2.563480	6.388318
	min	29.000000	35.000000	26.000000
	25%	34.000000	36.500000	27.750000
	50%	39.000000	38.000000	29.000000
	75%	40.000000	40.000000	39.250000
	max	41.000000	42.000000	43.000000
Income($)	**count**	3.000000	7.000000	12.000000
	mean	84000.000000	150000.000000	57291.666667
	std	5291.502622	12069.244660	7840.623622
	min	80000.000000	130000.000000	45000.000000
	25%	81000.000000	143500.000000	50625.000000
	50%	82000.000000	155000.000000	59000.000000
	75%	86000.000000	158000.000000	63250.000000
	max	90000.000000	162000.000000	70000.000000

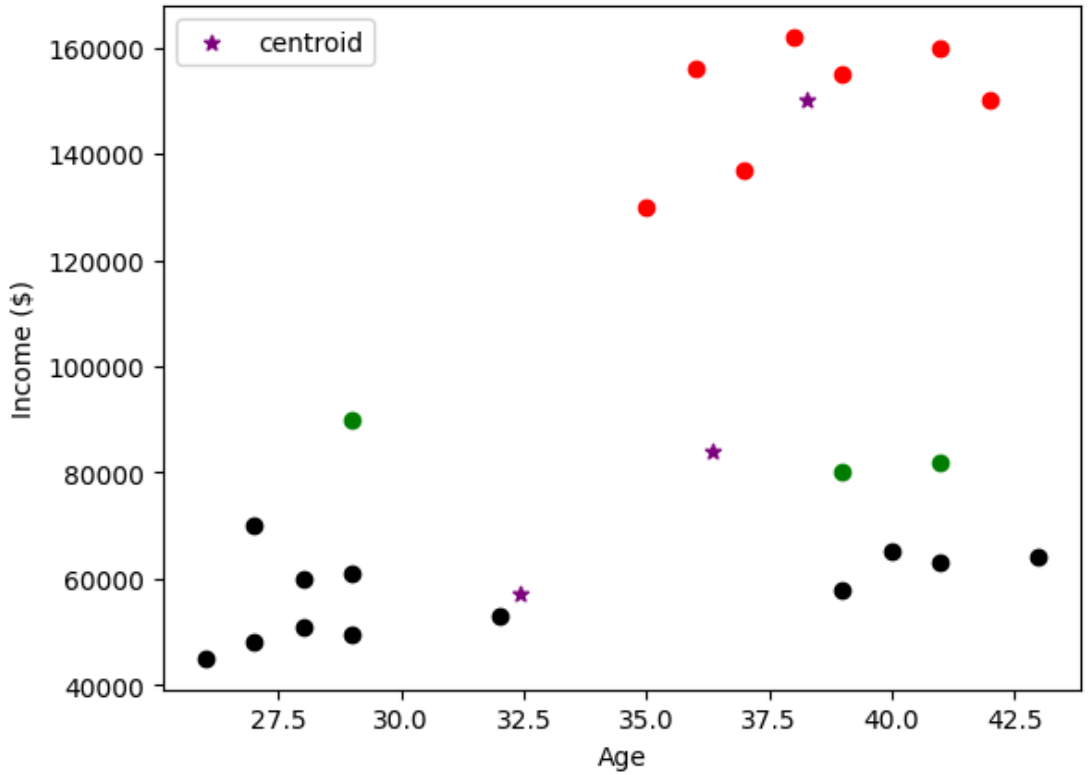
```
km.cluster_centers_
```

```
array([[3.63333333e+01, 8.40000000e+04],
       [3.82857143e+01, 1.50000000e+05],
       [3.24166667e+01, 5.72916667e+04]])
```

```
df1 = df[df.cluster==0]
df2 = df[df.cluster==1]
df3 = df[df.cluster==2]
plt.scatter(df1.Age,df1['Income($)'],color='green')
```

```
plt.scatter(df2.Age,df2['Income($)'],color='red')
plt.scatter(df3.Age,df3['Income($)'],color='black')
plt.scatter(km.cluster_centers_[:,0],km.cluster_centers_[:,1],color='purple',marker='*',label='centroid')
plt.xlabel('Age')
plt.ylabel('Income ($)')
plt.legend()
```

```
<matplotlib.legend.Legend at 0x7eb6bf063490>
```



Preprocessing using min max scaler

```
scaler = MinMaxScaler()

scaler.fit(df[['Income($)']])
df['Income($)'] = scaler.transform(df[['Income($)']])
```

```
5
6 scaler.fit(df[['Age']])
7 df['Age'] = scaler.transform(df[['Age']])
```

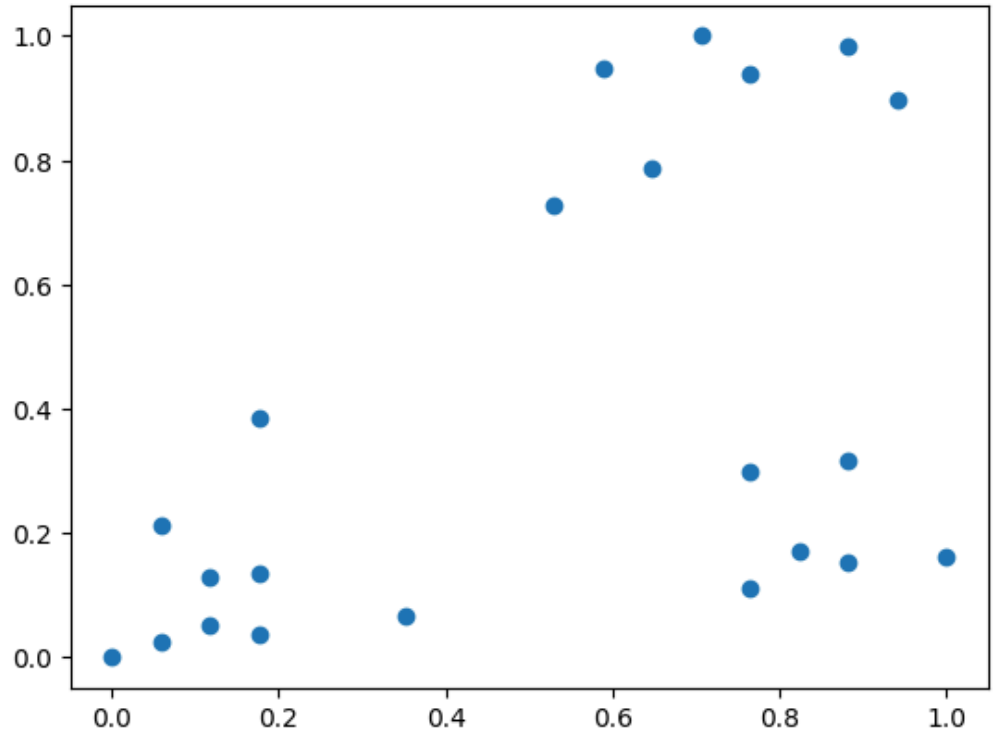
```
1 df.head()
```

	Name	Age	Income($)	cluster
0	Rob	0.058824	0.213675	2
1	Michael	0.176471	0.384615	0
2	Mohan	0.176471	0.136752	2
3	Ismail	0.117647	0.128205	2
4	Kory	0.941176	0.897436	1

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
1 plt.scatter(df.Age,df['Income($)'])
```

```
<matplotlib.collections.PathCollection at 0x7eb6bf0ccf90>
```




```
km = KMeans(n_clusters=3,init = 'k-means++')
y_predicted = km.fit_predict(df[['Age','Income($)']])
y_predicted
```

```
array([1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0],
      dtype=int32)
```

```
df['cluster']=y_predicted
df.head()
```

	Name	Age	Income($)	cluster
0	Rob	0.058824	0.213675	1
1	Michael	0.176471	0.384615	1
2	Mohan	0.176471	0.136752	1
3	Ismail	0.117647	0.128205	1
4	Kory	0.941176	0.897436	2

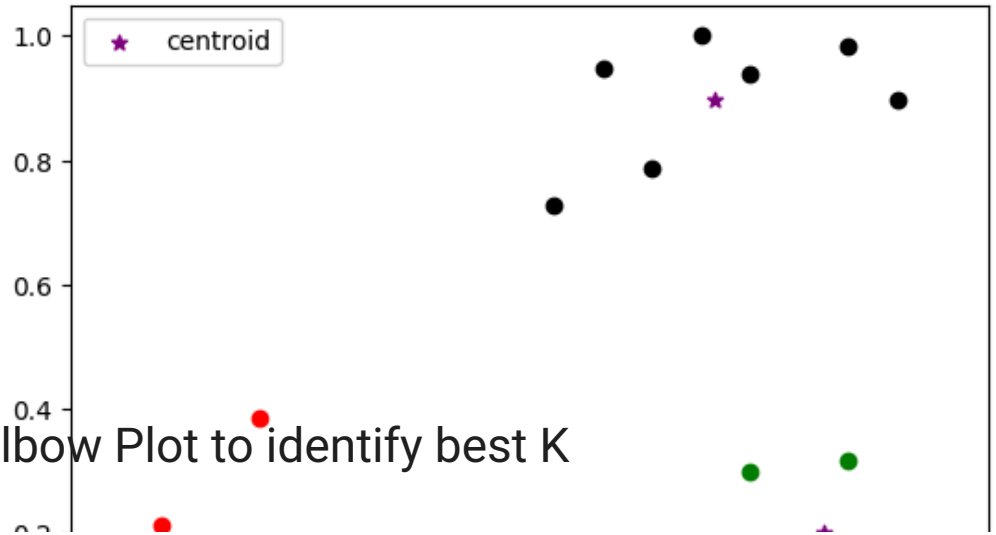
Next steps: Generate code with df | View recommended plots | New interactive sheet

```
km.cluster_centers_
```

```
array([[0.85294118, 0.2022792 ],
       [0.1372549 , 0.11633428],
       [0.72268908, 0.8974359 ]])
```

```
df1 = df[df.cluster==0]
df2 = df[df.cluster==1]
df3 = df[df.cluster==2]
plt.scatter(df1.Age,df1['Income($)'],color='green')
plt.scatter(df2.Age,df2['Income($)'],color='red')
plt.scatter(df3.Age,df3['Income($)'],color='black')
plt.scatter(km.cluster_centers_[:,0],km.cluster_centers_[:,1],color='purple',marker='*',label='centroid')
plt.legend()
```

```
<matplotlib.legend.Legend at 0x7eb6bf0eca10>
```



Elbow Plot to identify best K

```
wcss = []
k_rng = range(1,10)
for k in k_rng:
    km = KMeans(n_clusters=k, init='k-means++')
    km.fit(df[['Age','Income($)']])
    wcss.append(km.inertia_)
```

```
plt.xlabel('K')
plt.ylabel('wcss squared distance')
plt.plot(k_rng,wcss)
```

```
[<matplotlib.lines.Line2D at 0x7eb6bee2b610>]
```

